

Heme's relevance genuine? Re-visiting the roles of TANGO2 homologs including HRG-9 and HRG-10 in *C. elegans*

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eLife Assessment

This **useful** study provides **incomplete** evidence that TANGO2 homologs, including HRG-9 and HRG-10, are not heme chaperones but play a role in cellular bioenergetics and oxidative stress homeostasis. While outstanding strengths include the use of different model systems, genetic tools, and behavioral assays, there are weaknesses in the data presented for the conclusions drawn. Due to the differences in experimental protocols between this study and the previous work reported by Sun et al., it is insufficient to rule out the role of TANGO2 as a heme chaperone, and furthermore, the authors provide only indirect evidence for the role of TANGO2 in bioenergetic and oxidative stress pathways. Nevertheless, this study paves the way for future mechanistic studies addressing the mechanisms of how TANGO2 regulates oxidative stress independent of its previously demonstrated role as a heme chaperone.

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Abstract

Mutations in the *TANGO2* gene cause severe illness in humans, including life-threatening metabolic crises. However, the function of TANGO2 protein remains unknown. Using *Caenorhabditis elegans* and other models, it has recently been proposed that TANGO2 transports heme within and between cells, from areas with high heme concentrations to those with lower concentrations. Here, we demonstrate that heme-related observations in nematodes may be better explained by a previously unreported metabolic phenotype in these

worms, characterized by reduced feeding, decreased lifespan and brood sizes, and poor motility. We also show that several genes not implicated in heme transport are upregulated in the low heme state and conversely demonstrate that *hrg-9* in particular is highly responsive to oxidative stress, independent of heme status. Collectively, these data implicate bioenergetic failure and oxidative stress as key factors in the pathophysiology of TANGO2 deficiency, in alignment with observations from human patients. Our group performed several experiments in yeast and zebrafish deficient in TANGO2 homologs and was unable to replicate prior findings from these models. Overall, we believe there is insufficient evidence to support heme transport as the primary function for TANGO2.

Introduction

Transport and Golgi Organization 2 (TANGO2) Deficiency Disorder (TDD) is associated with a severe progressive neurodegenerative disease in children and young adults, frequently accompanied by recurrent, life-threatening metabolic crises.^{1,2,3,4} While TANGO2 is conserved in species across all three kingdoms of life, its cellular function has not been fully elucidated. Initially identified in a screen of *Drosophila* genes associated with ER-Golgi fusion,⁵ TANGO2 has since been shown to localize to both the cytoplasm and the mitochondria.⁶ Mutations in TANGO2 have been associated with altered mitochondrial respiration and morphology,⁷ dysregulated lipid homeostasis,^{8,9} and defects in fatty acid oxidation⁷ in multiple models. Treatment with supplemental pantothenic acid, the precursor of coenzyme A, has been shown to improve lipid profile abnormalities in cells,⁹ rescue phenotypic defects in a *Drosophila* TDD model,¹⁰ and potentially reduce metabolic crises in patients,¹¹ suggesting a central role for coenzyme A (CoA) and lipid homeostasis in the pathophysiology of TDD.

A recent study by Sun et al. reported that two *Caenorhabditis elegans* homologs of TANGO2, HRG-9 and HRG-10 (Heme-Responsive Gene), play a critical role in heme export from intestinal cells and lysosome-related organelles and that TANGO2 exhibits a similar function in mobilizing heme from mitochondria in other eukaryotic organisms including yeast.¹² Two subsequent studies also implicated TANGO2 in heme transport albeit via discordant intracellular mechanisms.^{13,14} The notion that defective heme transport underlies the pathophysiology of TANGO2 deficiency marks a sizeable shift from the emerging scientific consensus related to the role of TANGO2 in CoA and lipid handling. We hypothesize that these apparent defects in heme transport may be downstream features of aberrant cellular metabolism and may not be central to the pathophysiology of TDD.

Results

***C. elegans* lacking TANGO2 homologs (double knockout; DKO) demonstrate modest survival benefit upon toxic heme analog exposure**

We first sought to validate results from prior experiments using heme analogs in worms lacking both HRG-9 and HRG-10 (double knockout; DKO). In previous studies, DKO worms exposed to 1 μ M concentration of the toxic heme analog gallium protoporphyrin IX (GaPP) showed a significant survival benefit contrasted with the nearly uniform lethality in wild type (Bristol N2) worms.¹² We exposed gravid DKO and N2 nematodes to GaPP at varying concentrations, removing P_0 worms at 24 hours, and counting alive and dead F_1 progeny at 72 hours. While we required relatively higher concentrations of GaPP to achieve lethality, all strains exhibited a clear dose-dependent

reduction in survival (**Fig. 1A**). A statistically significant group effect was seen at the 2 μ M concentration with a relative survival benefit in DKO worms, though the difference between DKO and N2 worms was more modest than previously reported.

DKO *C. elegans* demonstrate increased lawn avoidance and reduced pharyngeal pumping

In maintaining the DKO strain under basal conditions (i.e., no GaPP exposure), we incidentally observed relatively intact *E. coli* lawns on plates housing DKO worms compared to N2 plates. On closer examination, DKO worms demonstrated several previously undescribed phenotypic features including lawn avoidance, a greater propensity for crawling off plates, reduced pharyngeal pumping, and decreased survival (**Fig. 2A, B, F**). Based on these observations, we hypothesized that the GaPP survival difference in our initial experiment might have been driven, at least in part, by reduced GaPP consumption. We therefore also examined the effect of GaPP exposure on worms missing a pharyngeal acetylcholine receptor subunit (*eat-2(ad465)*). The *eat-2* strain exhibits reduced pharyngeal pumping and has been used extensively as a model of dietary restriction.^{15,16} Of the three strains tested, survival was highest in *eat-2* mutants (**Fig. 1A**) despite the fact that these worms have no known defects in heme metabolism or transport.

DKO *C. elegans* demonstrate reduced ZnMP fluorescence

Sun et al. previously reported that DKO nematodes accumulate the fluorescent heme analog zinc mesoporphyrin IX (ZnMP) in intestinal cells after exposure in low-heme (4 μ M) liquid media. To determine whether this effect persisted under basal conditions on standard OP50 *E. coli* NGM plates, we exposed L4 nematodes to 40 μ M ZnMP and imaged them after 16 hours. Contrary to prior results, we found that DKO worms demonstrated significantly reduced intestinal ZnMP fluorescence compared to N2 (**Fig. 1B**). To test whether stress-induced gut granule autofluorescence secondary to heme deficiency may have contributed to the higher fluorescent intensities previously reported, we repeated this experiment under starved conditions and showed a marginal, albeit nonsignificant, increase in intestinal fluorescence. Given that this increase was comparable between DKO and N2 worms, we believe that autofluorescence is unlikely to account for their reported group difference.

DKO *C. elegans* demonstrate multiple features suggestive of bioenergetic dysfunction

In assessing brood survival in the GaPP assay, we also observed significantly smaller starting broods for DKO nematodes, a finding that persisted in the absence of GaPP (**Fig. 2C**). As oogenesis and egg-laying require a high energetic expenditure for gravid *C. elegans*, reduced brood size is a known feature of several metabolically impaired nematode strains.^{17,18} We also incidentally noticed reduced movement on the plate from DKO worms and decided to further characterize the worms' capacity for movement by subjecting them to a swim exhaustion assay. Nematodes were placed in isotonic M9 buffer and scored on their swimming ability at 4 minute intervals. While *C. elegans* are typically able to swim continuously for up to 90 minutes in M9 media,¹⁹ we observed that the DKO worms quickly became exhausted and could not maintain normal swimming shortly after being placed in media (**Fig. 2D**). We further quantified the rate of thrashing in M9 media during the first 4 minute interval and found that DKO worms thrashed 61% slower compared to N2s (**Fig. 2E**). Together, these results are suggestive of a potential bioenergetic defect arising from TANGO2 homolog deficiency.

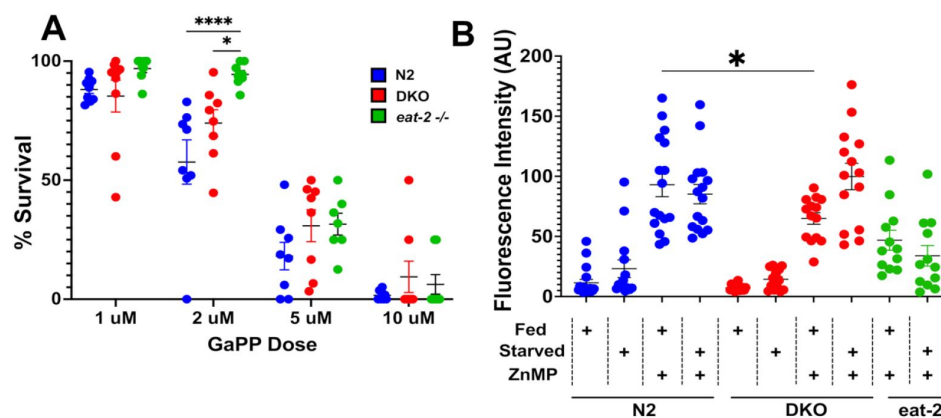


Figure 1

Reduced uptake of toxic and fluorescent heme analogs may explain marginal differences observed between wildtype *C. elegans* and nematodes lacking HRG-9 and HRG-10 (double knockout; DKO).

A) Survival of N2, DKO, or *eat-2* knockout worms exposed to 1, 2, 5, or 10 μ M GaPP for 72 hours. N=3 independent experiments. * $p < 0.05$, **** $p < 0.0001$ **B)** Quantification of fluorescent staining in N2, DKO, and *eat-2* worms under fed and starved conditions with or without 40 μ M ZnMP treatment. A.U. arbitrary units, N=15 worms analyzed over 3 independent experiments. * $p < 0.05$.

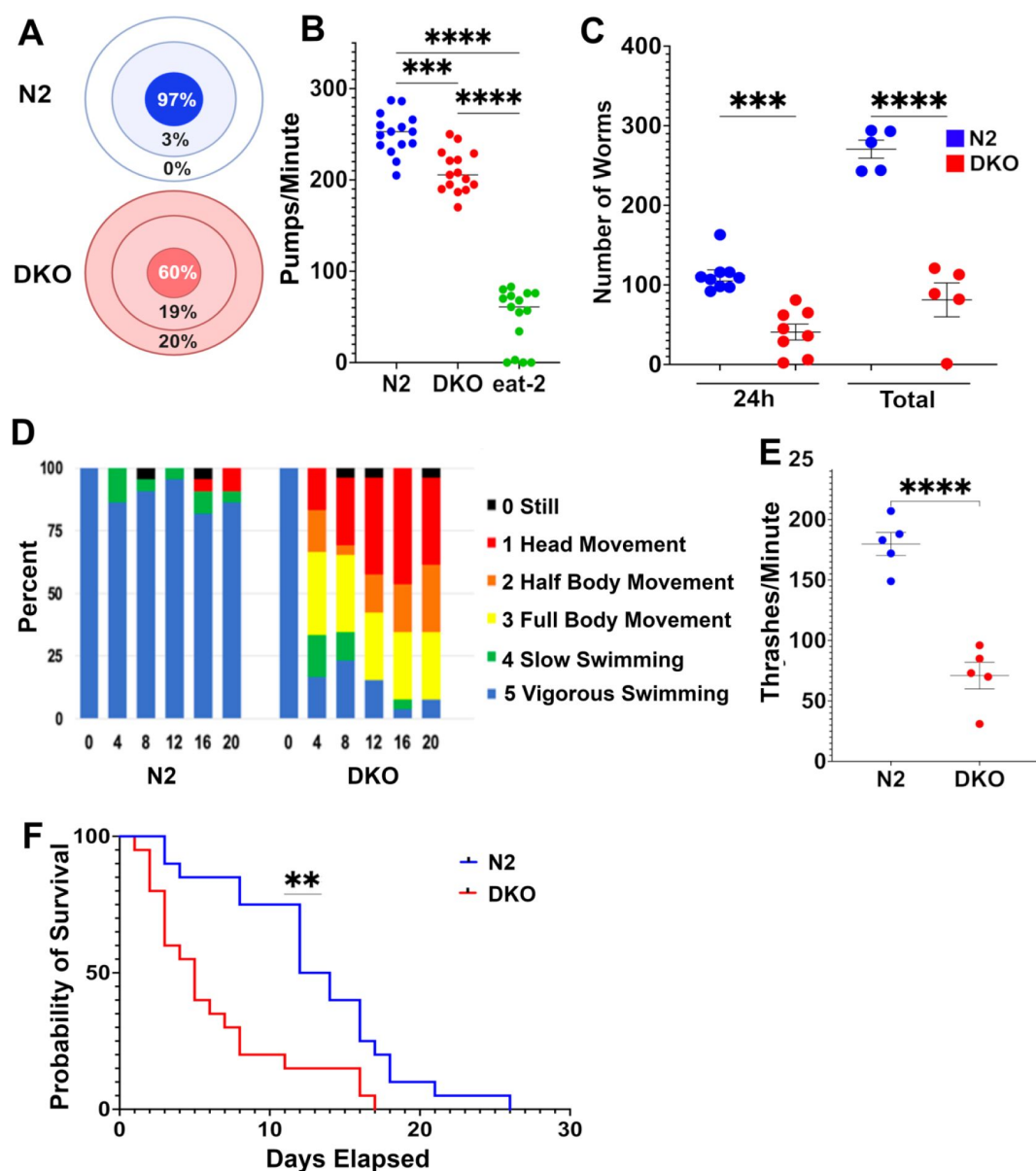


Figure 2

DKO nematodes demonstrate lawn avoidance and reduced pharyngeal pumping, brood sizes, motility, and survival.

A) Proportion of N2 or DKO worms present on OP50 lawn (innermost ring), off OP50 lawn (middle ring), or missing or dead from NGM plate (outer ring). N=180 worms over 3 independent experiments **B**) Number of pharyngeal pumps in a one-minute period in N2, DKO, and *eat-2* knockout worms. N=15 worms over 3 independent experiments **C**) Number of viable offspring laid by single adult N2 or DKO worms either after 24 hours of egg lay or across total egg-laying period (5 days). N=8 broods for 24-hour counts, N=5 broods for total brood size counts. **D**) Swimming behavior of N2 and DKO worms over a 20 minute interval. Worms were observed at 4 minute intervals and scored from 0-5 on swimming intensity. Bars represent the proportion of worms at each score. **E**) Quantification of thrashes after 4 minutes in M9 buffer. **F**) Longevity of N2 and DKO *C. elegans* observed from L4 larval stage. N=30 worms over 3 independent experiments. (** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$).

Oxidative stress is a transcriptional driver of TANGO2 homolog expression

Given previous observations implicating oxidative stress as a feature of TDD,⁷ we next sought to determine what other genes were enriched under low heme conditions. We reanalyzed the RNA-seq dataset generated by Sun et al., employing the Empirical Analysis of Gene Expression in R (edgeR) package on raw counts to accurately perform between-group comparisons across low (2 μ M), optimal (20 μ M), and high (400 μ M) heme conditions. We extracted the top 500 enriched genes and plotted those that showed significantly increased expression in the low heme state, based on computational clustering (N=134; **Fig. 3A**). Several genes with no known heme-related functions demonstrated stronger relative expression and higher likelihood ratios of conditional effect than did *hrg-9*. Furthermore, gene ontology analysis of genes with similar enrichment to *hrg-9* revealed a wide spectrum of biological processes and cellular roles, including but not limited to collagen deposition, cellular detoxification, and lipid binding (**Fig. 3B**). To test what alternate forms of cell stress might induce *hrg-9* and *hrg-10* expression, we exposed N2 nematodes to heat (34°C), starvation (24 hours without OP50), and paraquat, a potent generator of reactive oxygen species (25 mM), before performing RT-qPCR. We observed a 12-fold enrichment of *hrg-9* after paraquat exposure, suggesting that its expression may be linked to cellular stress more broadly and is not uniquely driven by heme levels (**Fig. 3C**).

Yeast deficient in TANGO2 homolog YGR127w exhibit normal growth

As Sun et al. also examined the function of TANGO2 homologs in yeast (YGR127w) and zebrafish (*tango2*), we sought to validate their findings in these models. Sun et al. reported a severe temperature sensitive growth defect and impaired heme distribution in the *ygr127w* yeast knockout. We also observed a growth defect in their strain. However, two separate strains on different backgrounds (BY4741 and BY4742) and a third strain generated by our lab (W303-1a) showed normal growth (**Fig. 4A**). The growth defect in their strain was also not rescued with YGR127w complementation. As the background used by Sun et al. is known to be prone to mitochondrial genome instability,²⁰ we hypothesize that their line may harbor a secondary mutation.

Myofiber defects in *tango2*-deficient zebrafish

In zebrafish, Sun et al. reported no discernable phenotype in *tango2*^{-/-} fish bred from heterozygous parents but observed severe skeletal muscle damage in *tango2*^{-/-} larvae from *tango2*^{-/-} parents. In a recent study, we showed increased lethality and reduced phospholipid and triglyceride levels in *tango2*^{-/-} fish obtained from heterozygous parents.²¹ *tango2*^{-/-} larvae from two different alleles (*tango2*^{bwh210} and *tango2*^{bwh211}) exhibited defects in myofiber organization irrespective of heterozygous or homozygous parents (**Fig. 4B**) but lacked the striking muscle damage reported by Sun et al.

Discussion

In this study, we attempted to replicate prior observations related to defective heme transport in a *C. elegans* model lacking TANGO2 homologs but were ultimately unable to do so. Additionally, we noted several previously unreported phenotypic characteristics that suggest a broader bioenergetic defect in these worms, reminiscent of what has been observed in patients with TDD. In our experiments with yeast and zebrafish, we were also unable to replicate central findings pertaining to growth and muscle fiber integrity, respectively.

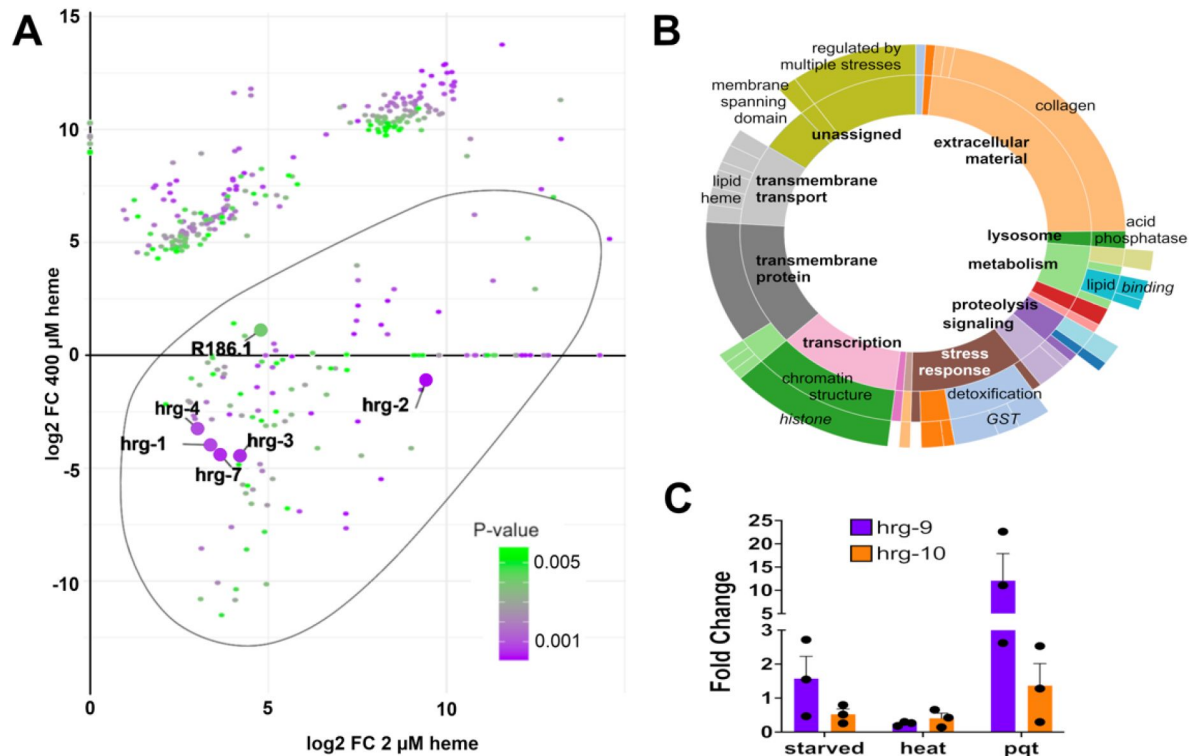


Figure 3

RNA-seq and qPCR analysis show that *hrg-9* and *hrg-10* are not uniquely heme responsive and may be linked to oxidative stress.

A) Analysis of top 500 genes with differential expression under low heme (2 μ M) and high heme (400 μ M) conditions. Outline represents 134 relevant genes identified by cluster analysis. **B)** Gene ontology analysis identified a variety of biological roles for genes within this cluster **C)** RT-qPCR of *hrg-9* and *hrg-10* under non-heme stress conditions: 24-hour starvation, 4-hour exposure to 34° C heat, and 25 mM paraquat. N=3 independent experiments.

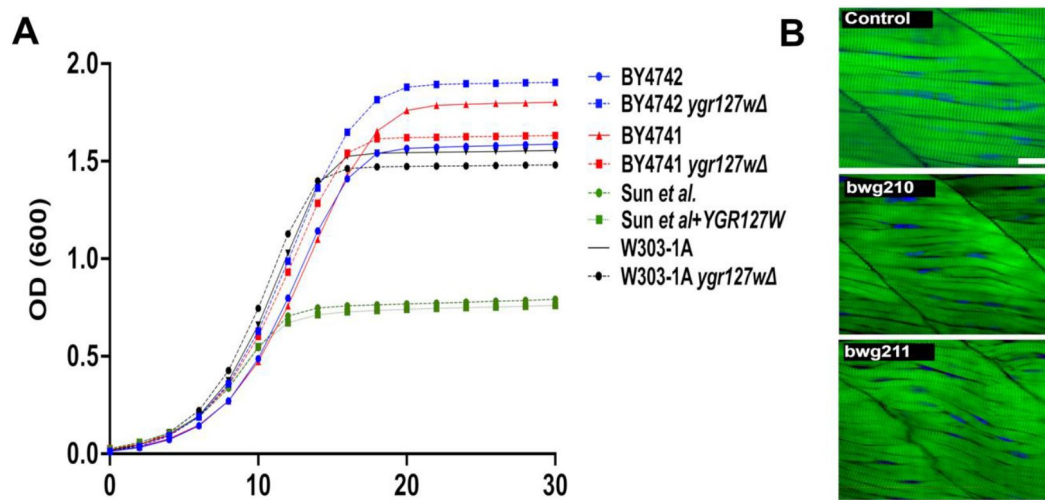


Figure 4

Growth and muscle fiber integrity in yeast and zebrafish models of TANGO2 deficiency fail to replicate previously reported phenotypes.

A) Yeast growth curves. Different strains were grown in SC medium or SC medium lacking histidine at 25° C. **B)** Whole mount phalloidin staining of control and two strains of *tango2*^{-/-} zebrafish (*bwg*²¹⁰ and *bwg*²¹¹). Myofibers in mutants lack the parallel organization observed in controls but do not demonstrate significant myofiber breakdown or damage. Representative images; N=8-10 in each group. Scale bar=5 μm.

Our data and collective clinical experience with TDD do not support the notion that TANGO2 primarily functions as a heme chaperone. Laboratory abnormalities in TDD include abnormal acylcarnitine profiles, hyperammonemia, and elevated creatinine kinase levels during metabolic crises,^{3,11} while abnormalities associated with defective heme transport (e.g., erythrocyte membrane defects, low hemoglobin levels) have not been linked with the disease. Strikingly, retrospective data suggest that patients with TDD receiving B-vitamin supplementation inclusive of pantothenic acid, a precursor of coenzyme A, do not experience metabolic crises¹¹ and show substantial improvement in other domains as well. Pantothenic acid supplementation also yielded full phenotypic rescue in a *Drosophila* model of the disease.¹⁰ It is difficult to reconcile how this treatment would be beneficial in a disease characterized by dysregulated heme trafficking.

Heme is a hydrophobic molecule, thus it is plausible that if TANGO2 and its homologs are involved in lipid handling, these proteins may also weakly bind heme.¹³ Han et al. demonstrated that a bacterial heme homolog, HtpA, directly binds heme and is necessary for cytochrome *c* function.¹³ We would note, however, that TANGO2 was not among the 378 heme-binding proteins identified on a recent proteomic screen of three separate cell lines.²² Jayaram et al. proposed that TANGO2 functions instead by interacting with FLVCR1b to release mitochondrial heme without directly binding to heme itself, though this interaction was observed only after heme synthesis was potentiated via d-ALA and iron supplementation, raising questions about the role of TANGO2 under basal conditions.¹⁴ Furthermore, in this study, GAPDH was identified as the exclusive binding partner of heme upon export through FLVCR1b. How GAPDH, a protein important for multiple cellular functions, including glycolysis, is affected in TDD remains unknown.

Although we believe there is currently insufficient evidence to suggest that TANGO2 is primarily involved in heme transport, more work is clearly needed to elucidate the cellular function of this important and highly conserved protein. This knowledge will be critical as we work to develop effective treatments for patients with TDD.

Materials and methods

Worm strains and maintenance

Worm strains used include Bristol N2 (obtained from Caenorhabditis Genetics Center (CGC) at the University of Minnesota), DKO (CCH303 (hrg-9(cck301)V; hrg-10(cck302)V) obtained from C. Chen, and eat-2(ad465), obtained from the Samuelson lab at the University of Rochester. All worms were maintained at 20° C on standard nematode growth medium (NGM) plates with OP50 *E. coli*. All *C. elegans* assays were performed and scored by blinded observers.

GaPP survival assay

Adult worms were placed on standard OP50 NGM plates treated with 1, 2, 5, or 10 μ M gallium protoporphyrin IX (Santa Cruz) and permitted to lay eggs for 24 hours then removed. Offspring were assessed 72 hours later for survival and were scored as dead if they did not respond to prodding with a platinum wire worm pick.

Pharyngeal pumping

L4 worms were observed and video recorded for 60 second intervals. The number of pharyngeal pumps per minute was manually counted.

Lawn avoidance

L4 worms were placed in the center of OP50 *E. coli* lawns on standard NGM plates and incubated at 20° C for 24 hours. Plates were examined 24 hours later and the numbers of worms remaining on the lawn or on unseeded agar were counted. Worms found on the sides of the dish or otherwise absent from NGM surface were scored as off the plate.

Brood size

For 24-hour brood assessments, young adult worms were placed on plates and allowed to lay eggs for 24 hours before being removed. Viable offspring were counted 48 hours later. For total brood assessments, L4 worms were placed on plates and moved to fresh plates every 24 hours for 5 days to ensure all eggs were accounted for. Viable offspring from eggs laid on each plate were counted 48 hours later and summed.

ZnMP fluorescence

L4 worms were placed on plates treated with 40 μ M Zinc mesoporphyrin IX (Santa Cruz) for 16 hours before being washed in M9 buffer, anesthetized in sodium azide, and mounted on slides with 2% agarose pads for imaging. Worms in the starved condition were placed on unseeded plates for 16 hours before preparation and mounting as above. Worms were imaged on an EVOS 5000 microscope with consistent acquisition parameters. Fluorescent intensity within the proximal worm intestine was measured in ImageJ.

RNA-seq cluster and gene ontology analysis

The RNA sequencing (RNA-seq) dataset generated by Sun et al. was analyzed with Empirical Analysis of Gene Expression in R (edgeR) and the top 500 genes were extracted. Computational cluster analysis was done in RStudio. Gene ontology analysis was performed on significantly enriched genes using WormCat.²³ Source code is available in supplemental materials.

Stress conditions and RT-qPCR

N2 worms were subjected to the following stress conditions prior to RNA extraction: 1) fasting: worms were deprived of OP50 *E. coli* for 24 hours; 2) heat: worms were incubated for 4 hours at 34° C; 3) paraquat: worms were placed on standard NGM/OP50 plates treated with 25 mM paraquat (methyl viologen; Sigma Aldrich) for 24 hours. Whole worm RNA was extracted with TriZol (Invitrogen) and treated with RNeasy Mini Kit (Qiagen). cDNA generation was performed with Maxima First Strand cDNA synthesis kit (Thermo Fisher). qPCR was performed in triplicate using iTaq Universal SYBR (Invitrogen) with a CFX Duet Realtime qPCR machine (BioRad). Gene expression was normalized to worm *act* and analyzed using the $\Delta\Delta C_q$ method. Primers used were as follows: *hrg-9*: GGACCCGCTGCCATACATAATC and GACAATTCAAATCTGGCATCGTG *hrg-10*: AGGCTTCCCGAGCACATTAC and CAGGCTCCATGCGTCTATCCAG *act*: CAACACTGTTCTTCCGGAG and CTTGATCTTCATGGTTGATGGG

Yeast strain construction

YGR127w was knocked out by homologous recombination. A strain (BY4741) harboring a *ygr127wD* was used as a template and the knockout cassette was amplified by PCR using the following primers that anneal upstream and downstream of the locus: TTGGCATCTGCCTAGCTTTTCG and AGCGTCTACTGTGGTTACTG The PCR amplicon was then transformed into W303-1a cells and transformants were selected on YPD plates containing 200mg/ml G418. Integration at the correct locus was confirmed by PCR using the same primers, as above. To complement the *ygr127wD* in the Sun et al. strain, wild type *YGR127w* was amplified with 400 base pairs on either side of the gene and cloned into a *HIS3*-containing plasmid (pRS413). Transformants were selected on synthetic complete (SC) medium lacking histidine.

Yeast growth curves

Cells were grown to stationary phase at 25°C in either SC or SC-histidine medium. Cultures in the same medium were inoculated at an OD600 of ~0.01 in a 100ml volume in a 96-well plate. The OD600 was read every 15 minutes on a Sunrise Tecan microplate reader.

Zebrafish

All procedures involving zebrafish were approved by the Brigham and Women's Hospital Animal Care and Use Committee. Fish were bred and maintained using standard methods as described.²⁴ tango2bwh210 and tango2bwh211 zebrafish lines were created in our laboratory by the CRISPR-Cas9 approach as described previously.²¹ The tango2bwg210 allele has an insertion of seven bases (c.226_227ins7; p.Tyr76Leufs*25) and tango2bwg211 has a 26 base insertions in exon2 (c.226_227ins26; p.Tyr76Leufs*207) resulting in frameshift mutations and loss of protein function. Whole mount phalloidin staining and microscopy was performed as described previously.²⁵

Statistics and Reproducibility

All statistical analyses were performed in GraphPad Prism 9. Data presented are mean ± S.E.M. One-way analysis of variance (ANOVA) followed by Bonferroni's multiple comparisons was used to determine statistical significance ($p < 0.05$). Sample sizes were not pre-determined.

Data availability statement

All raw data from worm behavior, yeast, zebrafish, and RNA-seq studies are available as a supplement to this paper.

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Additional information

Contributions

SJM, SES, KSY, APW, and LW designed the *C. elegans* experiments. SES, KSY, and MBB performed the *C. elegans* experiments and analyzed the data. LDO performed the RNA-seq analysis. MS designed the yeast experiments. PA performed the yeast experiments and analyzed the data. VAG and ESK designed the zebrafish experiments, and EK performed these experiments. VAG and ESK analyzed the zebrafish data. SJM and SES wrote the initial draft of the manuscript. All authors reviewed the manuscript and contributed to its final version.

Additional files

Extended data [↗](#)

R code [↗](#)

RNA seq output [↗](#)

Supplemental figure [↗](#)

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Reviewer #1 (Public review):

Summary:

Sandkuhler et al. re-evaluated the biological functions of TANGO2 homologs in *C. elegans*, yeast, and zebrafish. Compared to the previously reported role of TANGO2 homologs in transporting heme, Sandkuhler et al. expressed a different opinion on the biological functions of TANGO2 homologs. With the support of some results from their tests, they conclude that 'there is insufficient evidence to support heme transport as the primary function of TANGO2', in addition to their claims on the role of TANGO2 in modulating metabolism. While the differences are reported in this study, more work is needed to elucidate the biological function of TANGO2.

Strengths:

- (1) This work revisited a set of key experiments, including the toxic heme analog GaPP survival assay, the fluorescent ZnMP accumulation assay, and the multi-organismal investigations documented by Sun et al. in Nature 2022, which is critical for comparing the two works.
- (2) This work reported additional phenotypes for the *C. elegans* mutant of the TANGO2 homologs, including lawn avoidance, reduced pharyngeal pumping, smaller brood size, faster exhaustion under swimming test, and a shorter lifespan. These phenotypes are important for understanding the biological function of TANGO2 homologs, while they were missing from the report by Sun et al.
- (3) Investigating the 'reduced GaPP consumption' as a cause of increased resistance against the toxic GaPP for the TANGO2 homologs, *hrg-9 hrg-10* double null mutant provides a valuable perspective for studying the biological function of TANGO2 homologs.
- (4) This work thoroughly evaluated the role of TANGO2 homologs in supporting yeast growth using multiple yeast strains and also pointed out the mitochondrial genome instability feature of the yeast strain used by Sun et al.

Weaknesses:

- (1) A detailed comparison between this work and the work of Sun et al. on experimental protocols and reagents in the main text will be beneficial for readers to assess critically.
- (2) The GaPP used by Sun et al. (purchased from Frontier Scientific) is more effective in killing the worm than the one used in this study (purchased from Santa Cruz). Is the different outcome due to the differences in reagents? Moreover, Sun et al. examined the lethality after 3-4 days, while this work examined the lethality after 72 hours. Would the extra 24 hours make any difference in the result?
- (3) This work reported the opposite result of Sun et al. for the fluorescent ZnMP accumulation assay. However, the experimental protocols used by the two studies are massively different. Sun et al. did the ZnMP staining by incubating the L4-stage worms in an axenic mCeHR2 medium containing 40 μ M ZnMP (purchased from Frontier Scientific) and 4 μ M heme at 20 °C for 16 h, while this work placed the L4-stage worms on the OP50 *E. coli* seeded NGM plates treated with 40 μ M ZnMP (purchased from Santa Cruz) for 16 h. The liquid axenic mCeHR2 medium is bacteria-free, heme-free, and consistent for ZnMP uptake by worms. This work has mentioned that the *hrg-9 hrg-10* double null mutant has bacterial lawn avoidance and reduced pharyngeal pumping phenotypes. Therefore, the ZnMP staining protocol used in this work faces challenges in the environmental control for the wild type vs. the mutant. The authors should adopt the ZnMP staining protocol used by Sun et al. for a proper evaluation of fluorescent ZnMP accumulation.
- (4) A striking difference between the two studies is that Sun et al. emphasize the biochemical function of TANGO2 homologs in heme transporting with evidence from some biochemical

tests. In contrast, this work emphasizes the physiological function of TANGO2 homologs with evidence from multiple phenotypical observations. In the discussion part, the authors should address whether these observed phenotypes in this study can be due to the loss of heme transporting activities upon eliminating TANGO2 homologs. This action can improve the merit of academic debate and collaboration.

<https://doi.org/10.7554/eLife.105418.1.sa3>

Reviewer #2 (Public review):

Summary:

This work investigates the roles of TANGO2 orthologs in different model systems and suggests bioenergetic dysfunction and oxidative stress (and not heme metabolism) as crucial pathways in TANGO2 deficiency disorders (TDD). Specifically, studies in *C. elegans* showed that the lack of TANGO2 ortholog activity (i) does not provide a survival benefit upon toxic heme exposure; (ii) results in a series of defects related to energy levels (reduced pharyngeal pumping, lawn avoidance, poor motility, and low brood size); (iii) reduces the fluorescence of the heme analog ZnMP in the intestine. Furthermore, upon oxidative stress, one TANGO2 ortholog, *hrg-9*, is upregulated compared to control conditions. Additional studies on yeast and zebrafish models failed to replicate prior findings on heme distribution and muscle integrity.

These findings have a clear therapeutic impact, as TDD currently has no cure but only symptom-managing treatments. Identifying the correct pathway to correct the disease is pivotal to finding a cure.

Although compelling, the authors' primary claim is based on indirect evidence that only hints toward it. Unfortunately, I do not see any direct and convincing evidence linking TANGO2 orthologs to bioenergetic and oxidative stress pathways.

Strengths:

- (1) The study refutes and extends previous findings, highlighting new aspects of TANGO2's roles in cell physiology.
- (2) The use of different model systems to address the main research questions is useful.
- (3) The results suggest a broader impact than previously described, somewhat supporting the novelty of the study.

Weaknesses:

- (1) The manuscript is written mainly as a criticism of a previously published paper. Although reproducibility in science is an issue that needs to be acknowledged, a manuscript should focus on the new data and the experiments that can better prove and strengthen the new claims.
- (2) The current presentation of the logic of the study and its results does not help the authors deliver their message, although they possess great potential.
- (3) The study is missing experiments to link *hrg-9* and *hrg-10* more directly to bioenergetic and oxidative stress pathways.

<https://doi.org/10.7554/eLife.105418.1.sa2>

Reviewer #3 (Public review):

In this paper, Sandkuhler et al. reassessed the role of TANGO2 as a heme chaperone proposed by Sun et al in a recently published paper (<https://doi.org/10.1038/s41586-022-05347-z>) by partially repeating and failing to replicate experiments therein. Overall, Sandkuhler et al. conclude that the heme-related roles of TANGO2 had been overemphasized by Sun et al. especially because the *hrg9* gene does not exclusively respond to different regimens of heme synthesis/uptake but is susceptible to a greater extent to, for example, oxidative stress.

In recent years, the discussion around the heme-related roles of TANGO2 has been tantalizing but is still far from a definitive consensus. Discrepancies between results and their interpretation are a testament to how challenging and ambitious the understanding of TANGO2 and the phenotypes associated with TANGO2 defects are. Overall, the work presented by Sandkuhler et al. in this manuscript challenges the recent developments in the field and promotes the continuous characterisation of TANGO2 in relation to heme homeostasis.

A few comments and questions:

- (1) The authors stress - with evidence provided in this paper or indicated in the literature - that the primary role of TANGO2 and its homologues is unlikely to be related to heme trafficking, arguing that observed effects on heme transport are instead downstream consequences of aberrant cellular metabolism. But in light of a mounting body of evidence (referenced by the authors) connecting more or less directly TANGO2 to heme trafficking and mobilization, it is recommended that the authors comment on how they think TANGO2 could relate to and be essential for heme trafficking, albeit in a secondary, moonlighting capacity. This would highlight a seemingly common theme in emerging key players in intracellular heme trafficking, as it appears to be the case for GAPDH - with accumulating evidence of this glycolytic enzyme being critical for heme delivery to several downstream proteins.
- (2) The observation - using *eat-2* mutants and lawn avoidance behaviour - that survival patterns can be partially explained by reduced consumption, is fascinating. It would be interesting to quantify the two relative contributions.
- (3) In the legend to Figure 1A it's a bit unclear what the differently coloured dots represent for each condition. Repeated measurements, worms, independent experiments? The authors should clarify this.
- (4) It would help if the entire fluorescence images (raw and processed) for the ZnMP treatments were provided. Fluorescence images would also benefit Figure 1B.
- (5) Increasingly, the understanding of heme-dependent roles relies on transient or indirect binding to unsuspected partners, not necessarily relying on a tight affinity and outdated the notion of heme as a static cofactor. Despite impressive recent advancements in the detection of these interactions (for example <https://doi.org/10.1021/jacs.2c06104>; cited by the authors), a full characterisation of the hemome is still elusive. Sandkuhler et al. deemed it possible but seem to question that heme binding to TANGO2 occurs. However, Sun et al. convincingly showed and characterised TANGO2 binding to heme. It is recommended that the authors comment on this.

<https://doi.org/10.7554/eLife.105418.1.sa1>

Author response:

We have reviewed the helpful feedback from the reviewers and would like to thank them for their careful consideration of our manuscript. By way of provisional response, we agree with many of the above points and plan to revise our manuscript accordingly.

In an effort to replicate some of the heme trafficking-related experiments in the original paper using a *C. elegans* model of TDD, we were either unable to do so or demonstrated an alternative explanation for the findings we could partially reproduce. As the reviewers correctly point out, there were some methodological and reagent-related differences between the study by Sun et al. and our own that we will more directly highlight in a subsequent manuscript version. Additionally, where possible, we will attempt to replicate these experiments using the same protocol(s).

We observed several phenotypic traits observed in the *C. elegans* model of TDD that were not previously described in prior studies. While we believe these features to be consistent with a bioenergetic problem in the worm, direct evidence for this is admittedly lacking in our original manuscript. We are actively engaged in experiments examining potential functions of HRG-9 and HRG-10 unrelated to heme trafficking and will consider which data best aligns with the scope of this study, thus warranting inclusion in a subsequent manuscript version. We will also provide a more comprehensive review of relevant data generated by other groups (e.g., lipid dysregulation, impaired autophagy, mitochondrial dysfunction in the absence of TANGO2) in the discussion section.

Recommended improvements related to figure legends, terminology, and formatting will also be executed in our forthcoming version. On behalf of my co-authors and myself, thank you again for your time and effort improving this work.

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